DBMS – SALES AND CUSTOMER ANALYTICS SYSTEM

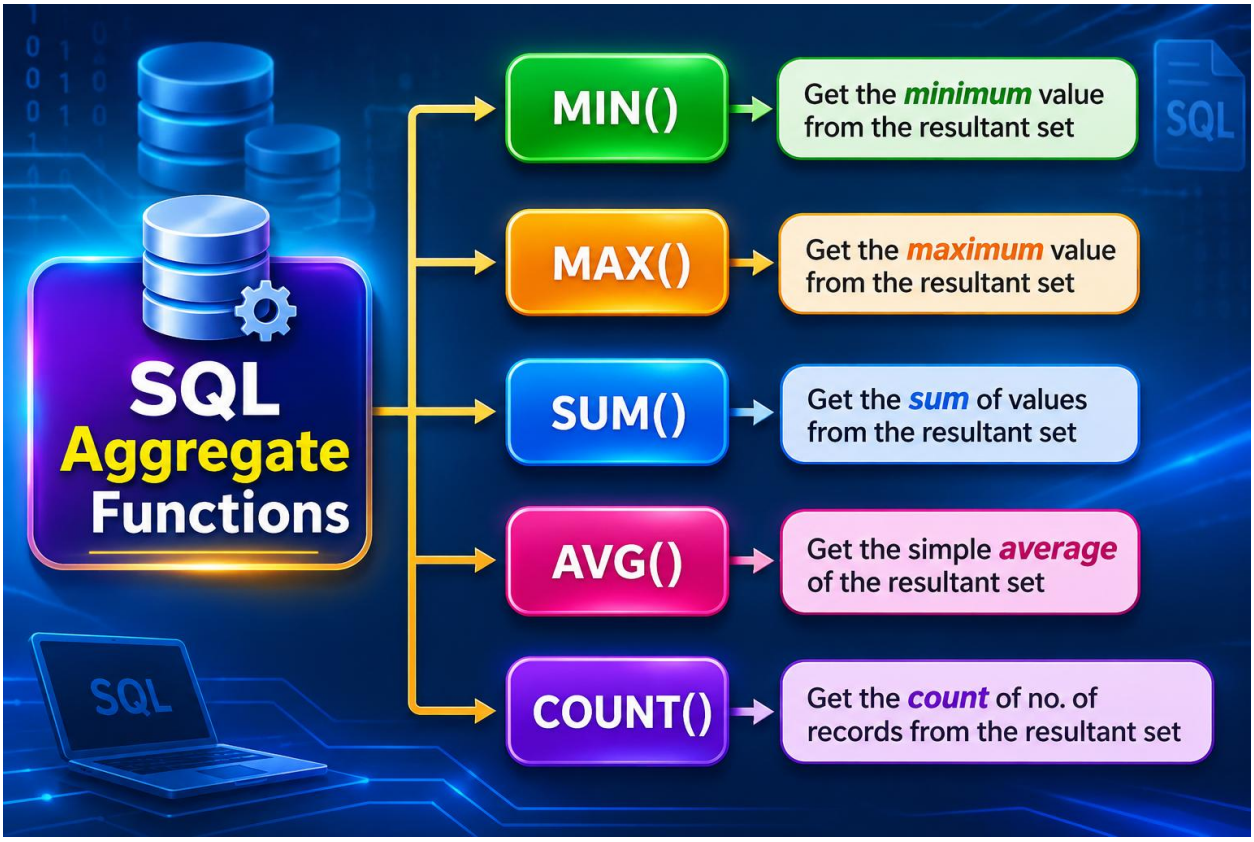
PROJECT TITLE: BOOKSWAGON - Online bookstore and online management system

WEEK 9

1. Aggregate Functions

SQL Aggregate Functions are used to perform calculations on a group of rows and return a single result.

Books wagon uses Aggregate Functions to analyses its online book sales by calculating the total orders, total sales revenue, average order value, and the highest and lowest order amounts.



1.1 COUNT() – Total number of orders

```
SQL> SELECT COUNT(Order_ID) AS Total_Orders
  2  FROM BookswagonOrders;

TOTAL_ORDERS
------------
          15
```

1.2 SUM()– Total sales amount

```
SQL> SELECT SUM(Total_Amount) AS Total_Sales
  2  FROM BookswagonOrders;

TOTAL_SALES
-----------
   11411.00
```

1.3 AVG() – Average order amount

```
SQL> SELECT AVG(Total_Amount) AS Average_Order_Amount
  2  FROM BookswagonOrders;

AVERAGE_ORDER_AMOUNT
--------------------
              760.73
```

1.4 MIN() – Lowest order amount

```
SQL> SELECT MIN(Total_Amount) AS Minimum_Order
  2  FROM BookswagonOrders;

MINIMUM_ORDER
-------------
       299.00
```

1.5 MAX() – Highest order amount

```
SQL> SELECT MAX(Total_Amount) AS Maximum_Order
  2  FROM BookswagonOrders;

MAXIMUM_ORDER
-------------
      1499.00
```

2. Generate Total Sales Report

```
SQL> SELECT COUNT(Order_ID) AS Total_Orders,
  2         SUM(Total_Amount) AS Total_Sales,
  3         AVG(Total_Amount) AS Average_Order_Value,
  4         MIN(Total_Amount) AS Minimum_Order,
  5         MAX(Total_Amount) AS Maximum_Order
  6  FROM BookswagonOrders;

TOTAL_ORDERS TOTAL_SALES AVERAGE_ORDER_VALUE MINIMUM_ORDER MAXIMUM_ORDER
------------ ----------- ------------------- ------------- -------------
          15    11411.00              760.73        299.00       1499.00
```

2.1 Customer-wise Sales Report

```
SQL> SELECT c.Customer_ID,
  2         c.Customer_Name,
  3         SUM(o.Total_Amount) AS Total_Purchase
  4  FROM BookswagonCustomer c
  5  INNER JOIN BookswagonOrders o
  6      ON c.Customer_ID = o.Customer_ID
  7  GROUP BY c.Customer_ID, c.Customer_Name
  8  ORDER BY Total_Purchase DESC;

CUSTOMER_ID CUSTOMER_NAME             TOTAL_PURCHASE
----------- ------------------------- --------------
          1 Priya                            1949.00
          5 Karthik S                        1298.00
          3 Arun Kumar                       1249.00
          2 Priya Kumar                      1248.00
          7 Rahul Kumar                      1197.00
          4 Divya Raj                        1074.00
          8 Keerthana R                       998.00
         10 Swetha Priya                      900.00
          9 Vignesh M                         749.00
          6 Harini Devi                       749.00

10 rows selected.
```

3. Find Top Customers Based on Purchase Amount

```
SQL> SELECT c.Customer_ID,
  2         c.Customer_Name,
  3         SUM(o.Total_Amount) AS Total_Purchase
  4  FROM BookswagonCustomer c
  5  INNER JOIN BookswagonOrders o
  6      ON c.Customer_ID = o.Customer_ID
  7  GROUP BY c.Customer_ID, c.Customer_Name
  8  ORDER BY Total_Purchase DESC;

CUSTOMER_ID CUSTOMER_NAME             TOTAL_PURCHASE
----------- ------------------------- --------------
          1 Priya                            1949.00
          5 Karthik S                        1298.00
          3 Arun Kumar                       1249.00
          2 Priya Kumar                      1248.00
          7 Rahul Kumar                      1197.00
          4 Divya Raj                        1074.00
          8 Keerthana R                       998.00
         10 Swetha Priya                      900.00
          9 Vignesh M                         749.00
          6 Harini Devi                       749.00

10 rows selected.
```

4. Identify Best-Selling Products

```
SQL> SELECT p.Product_ID,
  2         p.Product_Name,
  3         SUM(od.Quantity) AS Total_Quantity_Sold
  4  FROM BookswagonProduct p
  5  INNER JOIN BookswagonOrderDetails od
  6      ON p.Product_ID = od.Product_ID
  7  GROUP BY p.Product_ID, p.Product_Name
  8  ORDER BY Total_Quantity_Sold DESC;

PRODUCT_ID PRODUCT_NAME                       TOTAL_QUANTITY_SOLD
---------- ---------------------------------- -------------------
       101 The Alchemist                                        6
       104 Wings of Fire                                        1
       102 Atomic Habits                                        1
       103 The Psychology of Money                              1
       106 Rich Dad Poor Dad                                    1
```

4.1 Product Sales Amount

```
SQL> SELECT p.Product_ID,
  2         p.Product_Name,
  3         SUM(od.Quantity) AS Total_Quantity_Sold,
  4         SUM(od.Subtotal) AS Total_Sales
  5  FROM BookswagonProduct p
  6  INNER JOIN BookswagonOrderDetails od
  7      ON p.Product_ID = od.Product_ID
  8  GROUP BY p.Product_ID, p.Product_Name
  9  ORDER BY Total_Sales DESC;

PRODUCT_ID PRODUCT_NAME                       TOTAL_QUANTITY_SOLD TOTAL_SALES
---------- ---------------------------------- ------------------- -----------
       101 The Alchemist                                        6     2700.00
       102 Atomic Habits                                        1      499.00
       106 Rich Dad Poor Dad                                    1      399.00
       103 The Psychology of Money                              1      350.00
       104 Wings of Fire                                        1      299.00
```

5. Category-wise Sales Analysis

```
SQL> SELECT cat.Category_ID,
  2         cat.Category_Name,
  3         SUM(od.Quantity) AS Total_Quantity_Sold,
  4         SUM(od.Subtotal) AS Total_Sales
  5  FROM BookswagonCategory cat
  6  INNER JOIN BookswagonProduct p
  7      ON cat.Category_ID = p.Category_ID
  8  INNER JOIN BookswagonOrderDetails od
  9      ON p.Product_ID = od.Product_ID
 10  GROUP BY cat.Category_ID, cat.Category_Name
 11  ORDER BY Total_Sales DESC;

CATEGORY_ID CATEGORY_NAME             TOTAL_QUANTITY_SOLD TOTAL_SALES
----------- ------------------------- ------------------- -----------
          1 Fiction                                     7     3199.00
          2 Finance                                     2      749.00
          3 Biography                                   1      299.00
```

6. Combined Business Sales Report

```
SQL> SELECT c.Customer_Name,
  2         o.Order_ID,
  3         TO_CHAR(o.Order_Date, 'DD-MON-YYYY') AS Order_Date,
  4         p.Product_Name,
  5         cat.Category_Name,
  6         od.Quantity,
  7         od.Subtotal
  8  FROM BookswagonCustomer c
  9  INNER JOIN BookswagonOrders o
 10      ON c.Customer_ID = o.Customer_ID
 11  INNER JOIN BookswagonOrderDetails od
 12      ON o.Order_ID = od.Order_ID
 13  INNER JOIN BookswagonProduct p
 14      ON od.Product_ID = p.Product_ID
 15  INNER JOIN BookswagonCategory cat
 16      ON p.Category_ID = cat.Category_ID
 17  ORDER BY o.Order_ID;
```

CUSTOMER_NAME	ORDER_ID	ORDER_DATE	PRODUCT_NAME	CATEGORY_NAME	QUANTITY	SUBTOTAL
Priya	1001	18-OCT-2026	The Alchemist	Fiction	1	450.00
Priya Kumar	1002	19-OCT-2026	The Alchemist	Fiction	1	450.00
Priya Kumar	1002	19-OCT-2026	Atomic Habits	Fiction	1	499.00
Arun Kumar	1003	20-OCT-2026	The Psychology of Money	Finance	1	350.00
Arun Kumar	1003	20-OCT-2026	The Alchemist	Fiction	1	450.00
Divya Raj	1004	21-OCT-2026	Wings of Fire	Biography	1	299.00
Divya Raj	1004	21-OCT-2026	The Alchemist	Fiction	1	450.00
Karthik S	1005	22-OCT-2026	Rich Dad Poor Dad	Finance	1	399.00
Karthik S	1005	22-OCT-2026	The Alchemist	Fiction	1	450.00
Priya	5001	18-SEP-2026	The Alchemist	Fiction	1	450.00

10 rows selected.